to us from the shadow's left edge move in the same direction as the hole's whirling space. The space whirl gives them a boost, letting them escape from closer to the horizon than light rays on the shadow's right edge, which struggle against the whirl of space. That's why the shadow is flattened on the left and bulges out on the right. In Chapter 8 I talk more about this and other aspects of what a black hole really looks like, as seen up close in our universe, in our *brane*.

How Do We Know This Is *True*?

Einstein's relativistic laws have been tested to high precision. I'm convinced they are right, except when they confront quantum physics. For a big black hole like *Interstellar*'s Gargantua, quantum physics is relevant only near its center, in its singularity. So if black holes exist at all in our universe, they must have the properties that Einstein's relativistic laws dictate, the properties I described above.

These properties and others have been deduced from Einstein's equations by a large number of physicists standing intellectually on each others' shoulders (Figure 5.7); most importantly, Karl Schwarzschild, Roy Kerr, and Stephen Hawking. In 1915, shortly before his tragic death on World War I's German/Russian front, Schwarzschild deduced the details of the warped spacetime around a nonspinning black hole. In physicists' jargon, those details are called the "Schwarzschild metric." In 1963, Kerr (a New Zealand mathematician) did the same for a spinning black hole: he deduced the spinning hole's "Kerr metric." And in the early 1970s Stephen Hawking and others deduced a set of laws that black holes must obey when they swallow stars, collide and merge, and feel the tidal forces of other objects.

Black holes surely do exist. Einstein's relativistic laws insist that, when a massive star exhausts the nuclear fuel that keeps it hot, then the